

University of Utah

Fall 2025

MATH 3220-002

Midterm 2 Questions

Instructor: Alp Uzman

October 31, 2025, 9:40 AM - 10:30 AM

Surname:

First Name:

uNID:

KEY

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1. [47 points] There is nothing to prove in this problem. Consider the function $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by

$$(r, \theta) \mapsto (r \cos(\theta), r \sin(\theta)).$$

Perform the following tasks:

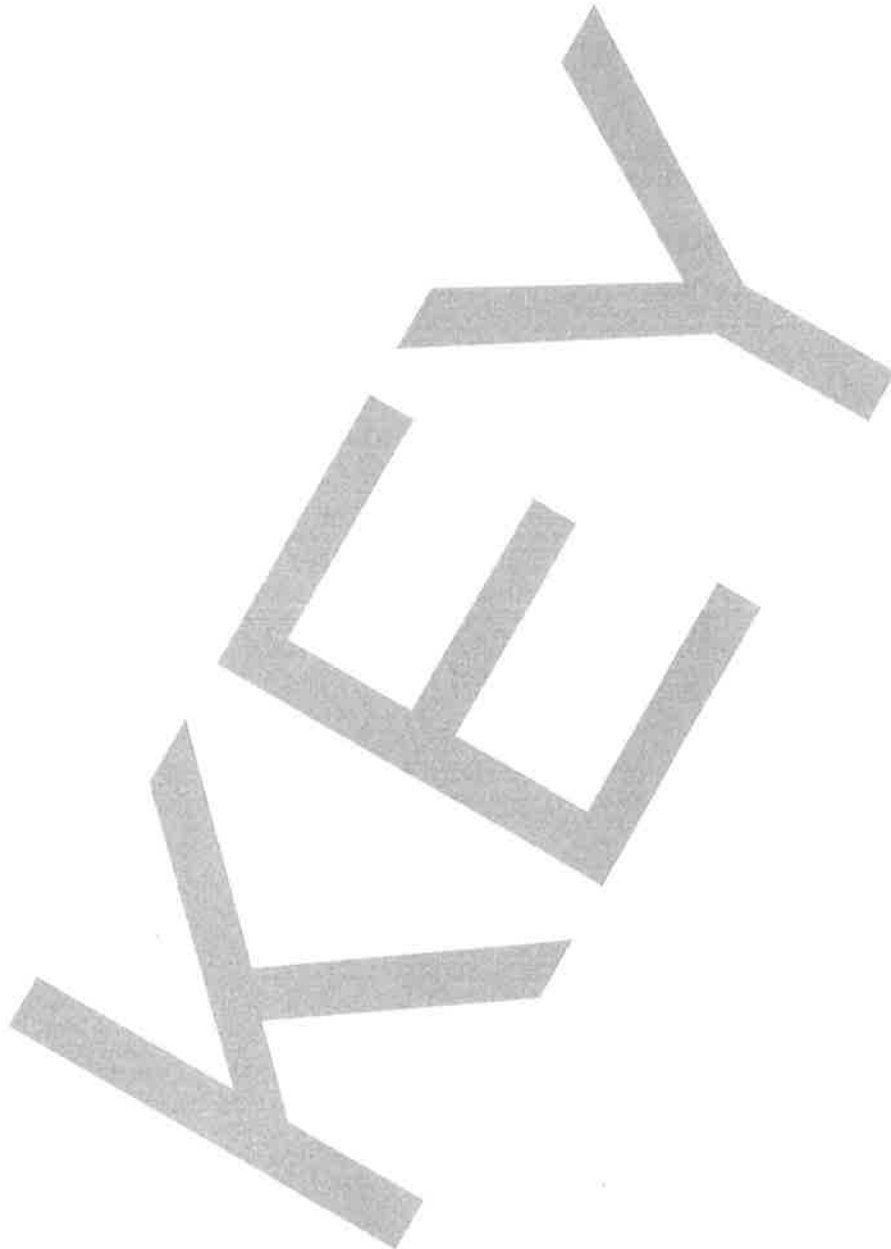
- (a) [20 points] Compute the derivative of f at $(1, \pi/2)$.

$$f'(r, \theta) = \begin{pmatrix} \frac{\partial f_1}{\partial r}(r, \theta) & \frac{\partial f_1}{\partial \theta}(r, \theta) \\ \frac{\partial f_2}{\partial r}(r, \theta) & \frac{\partial f_2}{\partial \theta}(r, \theta) \end{pmatrix}$$

$$= \begin{pmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{pmatrix}$$

$$\Rightarrow \boxed{f'\left(1, \frac{\pi}{2}\right) = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}}$$

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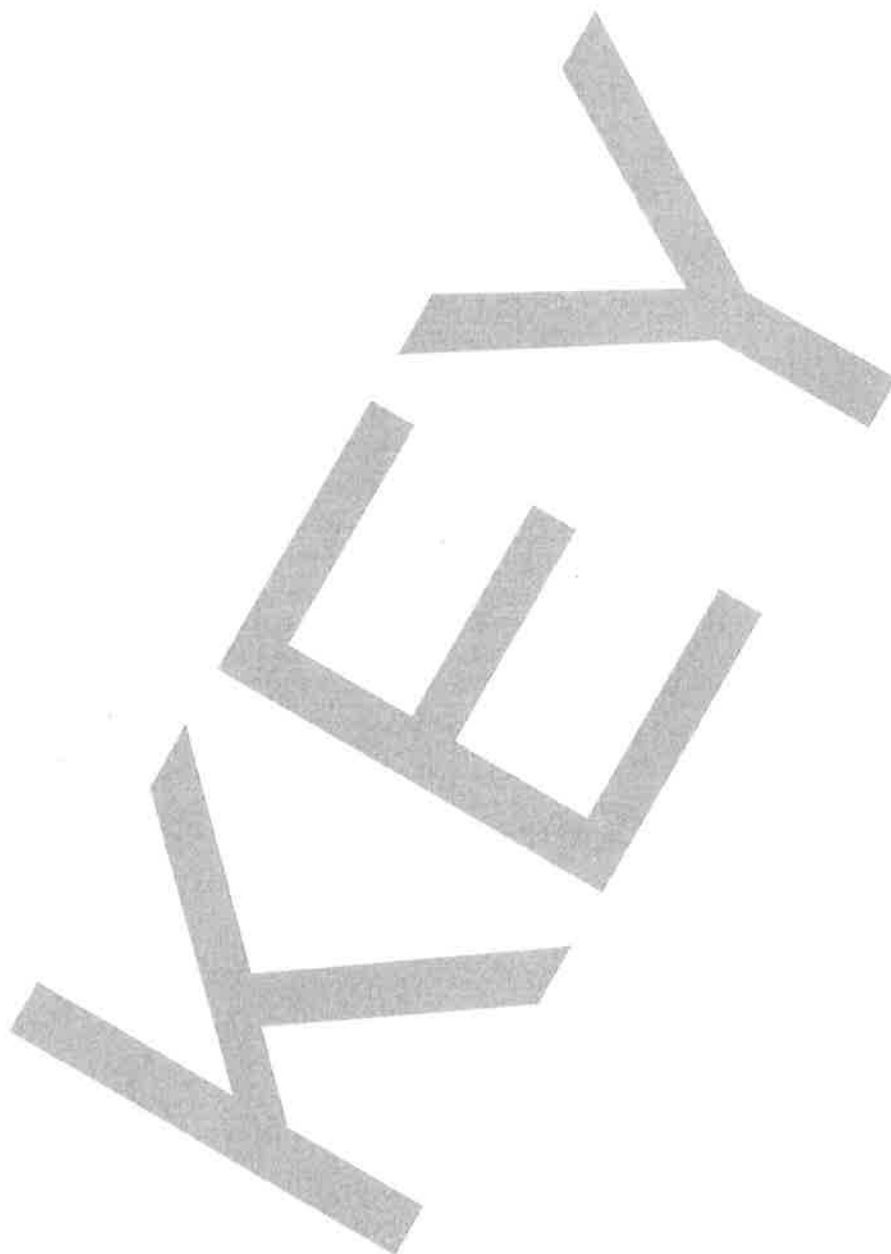
- (b) [20 points] Compute the directional derivative of f at $(1, \pi/2)$ in the direction of $(0, 1)$.

$$D_{(0,1)} f(1, \pi/2) = f'(1, \pi/2) \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$\stackrel{(a)}{=} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \end{pmatrix}$$

$$\Rightarrow \boxed{D_{(0,1)} f(1, \pi/2) = (-1, 0)}$$

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(c) [3 points] Compute the second derivative of f at $(1, \pi/2)$.

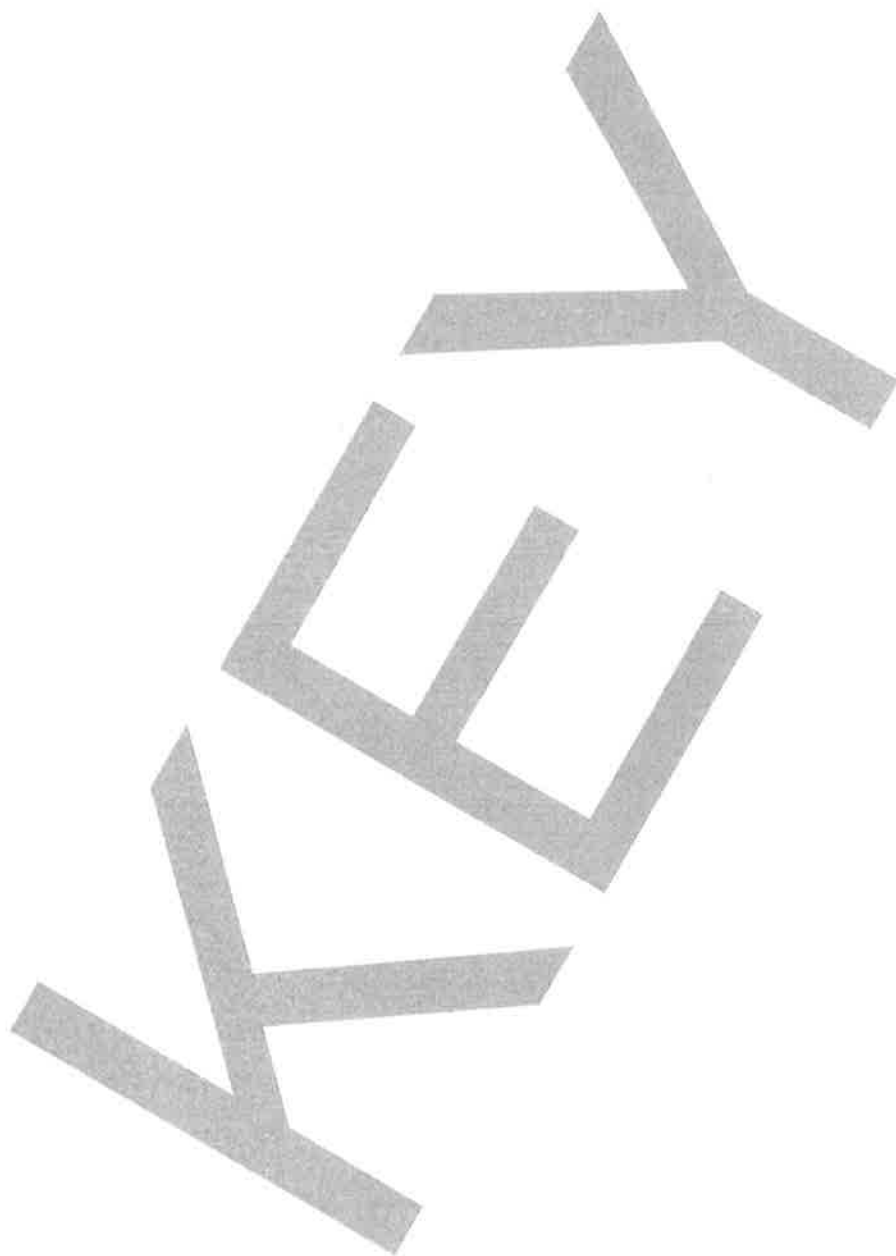
$$f''(r, \theta) \left((h_1, h_2), (k_1, k_2) \right)$$

$$= \begin{pmatrix} h_1 \\ h_2 \end{pmatrix}^T \begin{pmatrix} \frac{\partial^2 f_1}{\partial r^2} & \frac{\partial^2 f_1}{\partial r \partial \theta} \\ \frac{\partial^2 f_1}{\partial r \partial \theta} & \frac{\partial^2 f_1}{\partial \theta^2} \\ \frac{\partial^2 f_2}{\partial r^2} & \frac{\partial^2 f_2}{\partial r \partial \theta} \\ \frac{\partial^2 f_2}{\partial r \partial \theta} & \frac{\partial^2 f_2}{\partial \theta^2} \end{pmatrix} \begin{pmatrix} k_1 \\ k_2 \end{pmatrix}$$

$$= \begin{pmatrix} h_1 \\ h_2 \end{pmatrix}^T \begin{pmatrix} 0 & -\sin \theta \\ -\sin \theta & -r \cos \theta \\ 0 & \cos \theta \\ \cos \theta & -r \sin \theta \end{pmatrix} \begin{pmatrix} k_1 \\ k_2 \end{pmatrix}$$

$$\Rightarrow f''(1, \pi/2) = \begin{pmatrix} 0 & -1 \\ -1 & 0 \\ 0 & 0 \\ 0 & -1 \end{pmatrix}$$

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(d) [2 points] Compute the order 2 Taylor polynomial of f at $(1, \pi/2)$.

$$f(1, \pi/2) = (0, 1) = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

$$f'(1, \pi/2)(h_1, h_2) = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} h_1 \\ h_2 \end{pmatrix} = \begin{pmatrix} -h_2 \\ h_1 \end{pmatrix}.$$

$$f''(1, \pi/2)(h_1, h_2)^2 = \begin{pmatrix} h_1 \\ h_2 \end{pmatrix}^T \begin{pmatrix} 0 & -1 \\ -1 & 0 \\ 0 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} h_1 \\ h_2 \end{pmatrix}$$

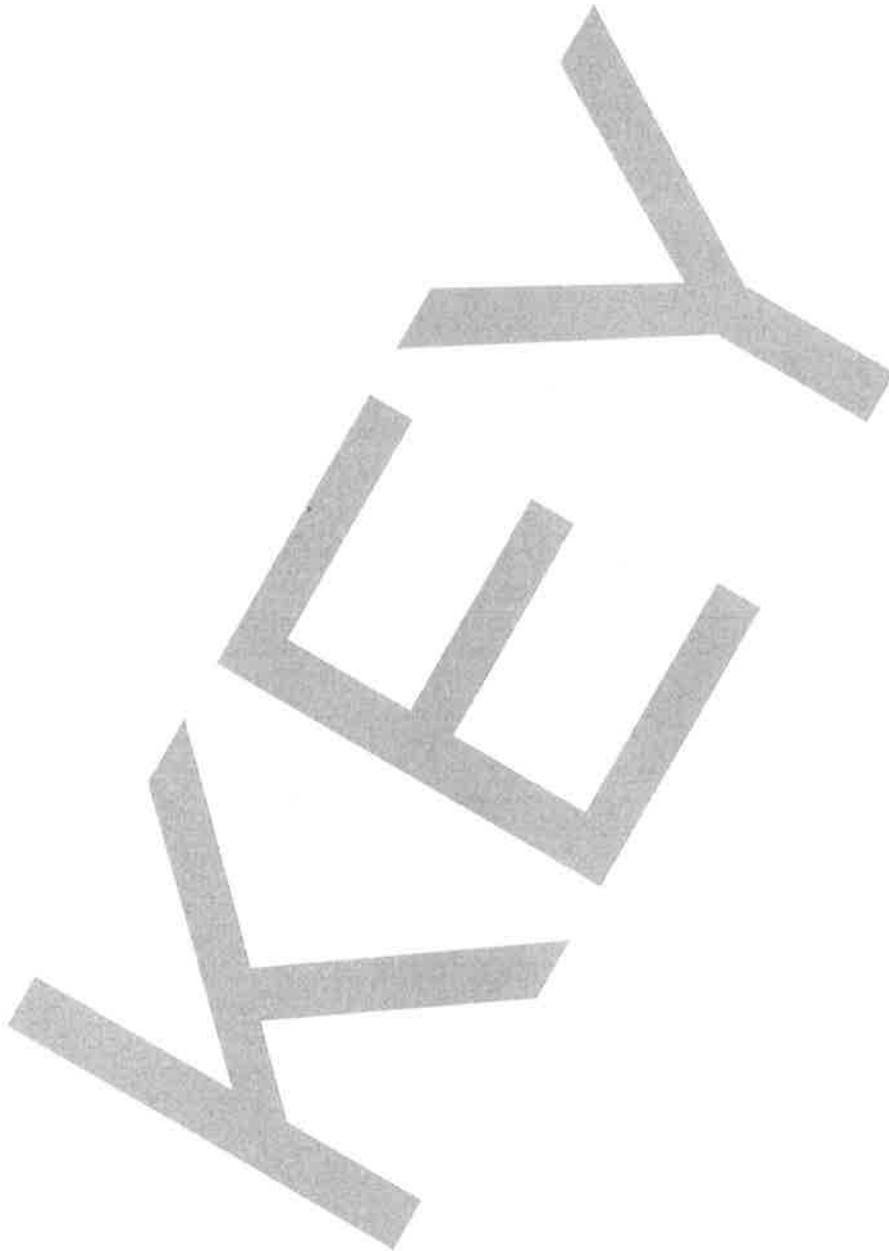
$$= \begin{pmatrix} -2h_1 h_2 \\ -h_2^2 \end{pmatrix}$$

$$\begin{aligned} &\Rightarrow T_2(f; (1, \pi/2))(1+h_1, \pi/2+h_2) \\ &= f(1, \pi/2) + f'(1, \pi/2)(h_1, h_2) + \frac{1}{2} f''(1, \pi/2)(h_1, h_2)^2 \\ &= \begin{pmatrix} 0 \\ 1 \end{pmatrix} + \begin{pmatrix} -h_2 \\ h_1 \end{pmatrix} + \frac{1}{2} \begin{pmatrix} -2h_1 h_2 \\ -h_2^2 \end{pmatrix} = \begin{pmatrix} -(1+h_1)h_2 \\ 1+h_1 - \frac{1}{2}h_2^2 \end{pmatrix} \end{aligned}$$

$$\bar{r} = 1+h_1, \quad \bar{\theta} = \frac{\pi}{2}+h_2.$$

$$\Rightarrow T_2(f; (1, \pi/2))(\bar{r}, \bar{\theta}) = \begin{pmatrix} -\bar{r} (\bar{\theta} - \frac{\pi}{2}) \\ \bar{r} - \frac{1}{2} (\bar{\theta} - \frac{\pi}{2})^2 \end{pmatrix}.$$

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- (e) [2 points] Find all points (r, θ) such that f is a local diffeomorphism near (r, θ) .

By Inverse Function Theorem,
 f is a local diffeo. near (r, θ)

$$\Leftrightarrow f'(r, \theta) \text{ inv.}$$

$$\Leftrightarrow \det(f'(r, \theta)) \neq 0.$$

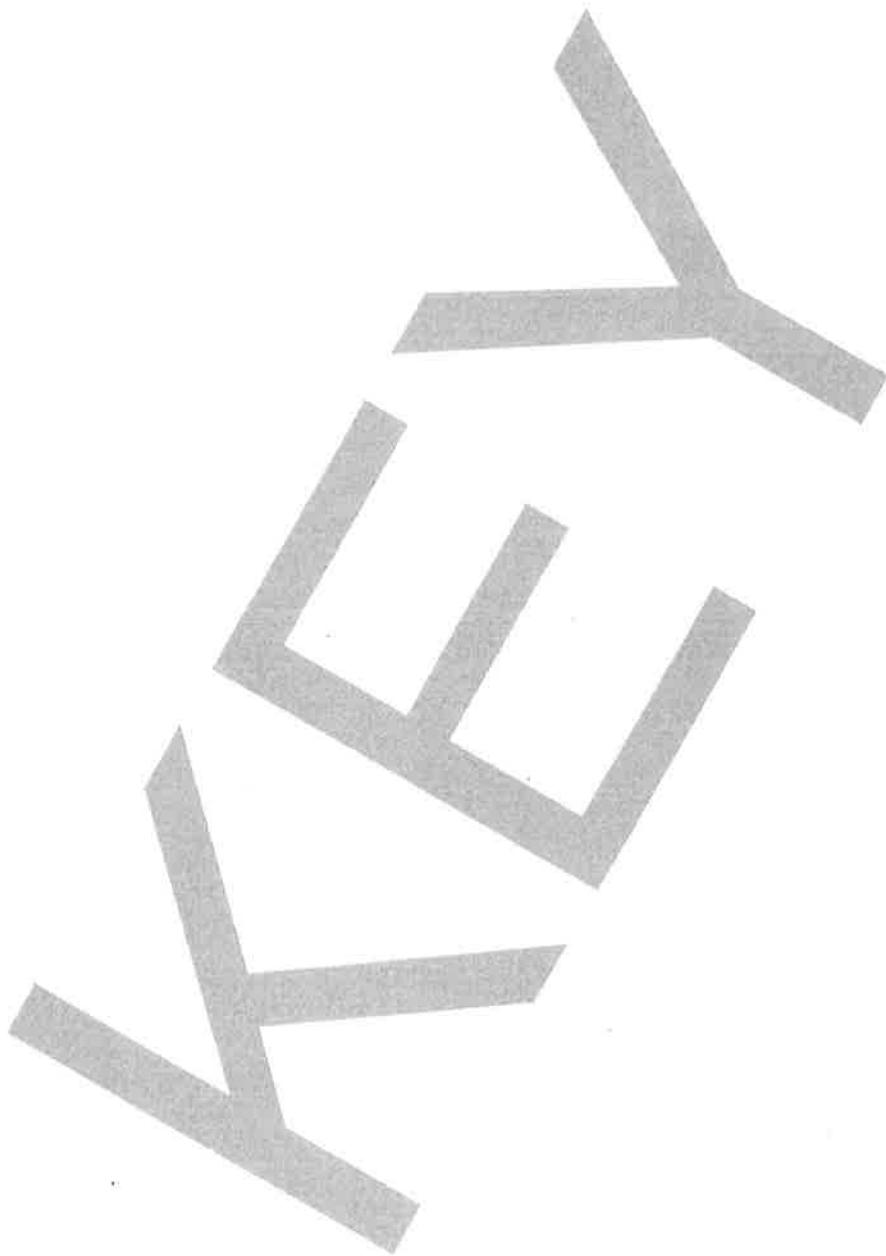
$$\Leftrightarrow 0 \neq \det \begin{pmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{pmatrix}$$

$$= r \cos^2 \theta + r \sin^2 \theta$$

$$= r$$

$$\Leftrightarrow \boxed{r \neq 0}$$

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2. [50 points] Let M be an open subset of $\mathbb{R}^2$, and let $f : M \rightarrow \mathbb{R}$ be a continuously differentiable function. Consider the following properties:

A: $\exists \phi \in C^1(\mathbb{R}; \mathbb{R}), \forall (x, y) \in M : f(x, y) = \phi(y).$

(In words, f is constant in the first variable.)

B: $\forall (x, y) \in M : \frac{\partial f}{\partial x}(x, y) = 0.$

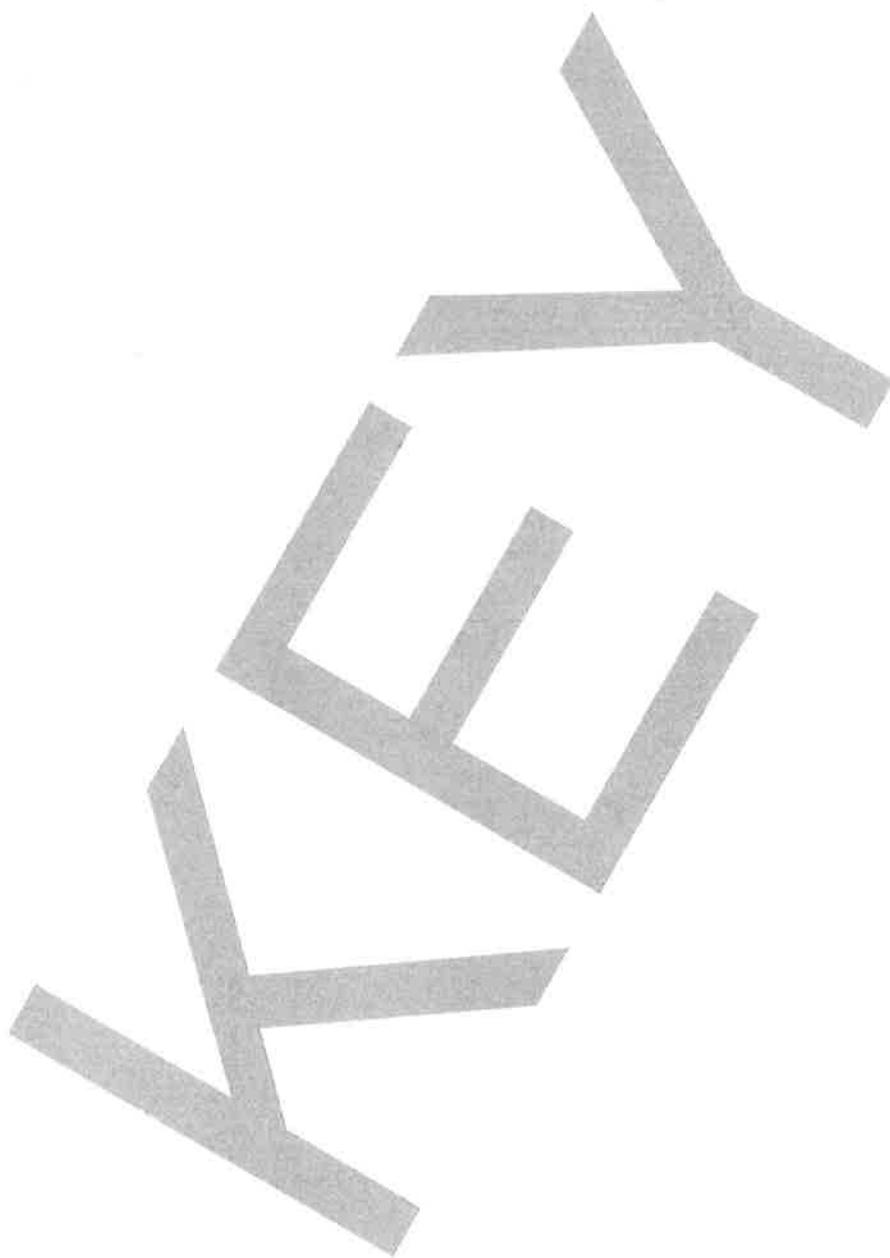
(In words, the x -partial of f vanishes.)

- (a) [25 points] **Prove or disprove the following statement:**

If f satisfies property **A**, then it must satisfy property **B**.

$$\begin{aligned} \frac{\partial f}{\partial x}(x, y) &= \lim_{h \rightarrow 0} \frac{f(x+h, y) - f(x, y)}{h} \\ &= \lim_{h \rightarrow 0} \frac{\phi(y) - \phi(y)}{h} = 0 \quad \checkmark \end{aligned}$$

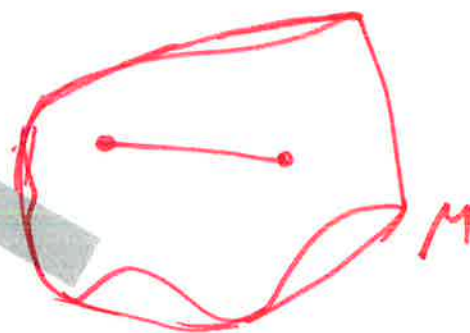
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- (b) [15 points] Prove or disprove the following statement:
 Suppose M is convex. If f satisfies property **B**, then it must satisfy property **A**.

Let $(x_1, y_1), (x_2, y_2) \in M$, ~~$y_1 \neq y_2$~~
 be such that $y_1 = y_2$.

Then by MVT applied
 to these two points,



$$|f(x_1, y_1) - f(x_2, y_2)|$$

$$\leq \left(\sup_{0 \leq t \leq 1} \left| \frac{\partial f}{\partial x}((1-t)x_1 + tx_2, y_2) \right| \right) |x_2 - x_1|$$

$$= 0$$

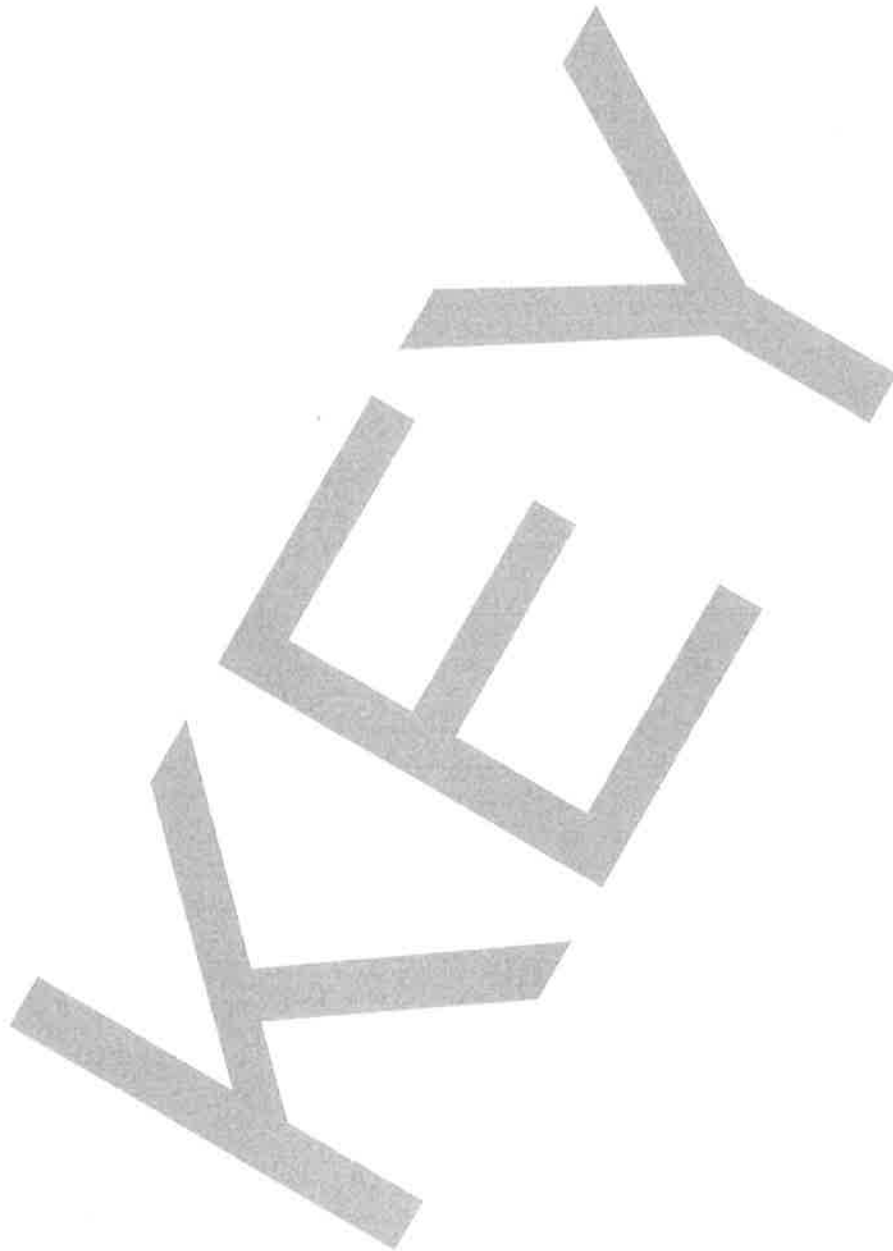
$$\Rightarrow f(x_1, y_1) = f(x_2, y_2).$$

Then $\phi: \mathbb{R} \rightarrow \mathbb{R}$

$y \mapsto f(x_1, y)$ is well defined

□

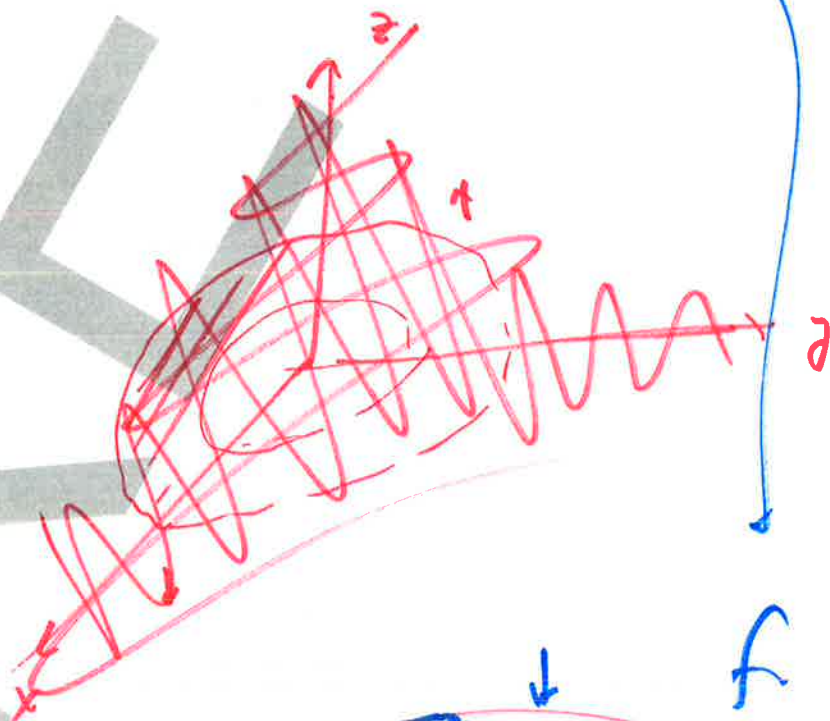
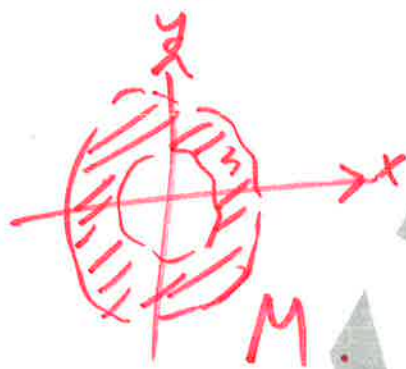
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- (c) [10 points] **Prove or disprove the following statement:**
 Suppose M is path connected. If f satisfies property **B**, then
 it must satisfy property **A**.

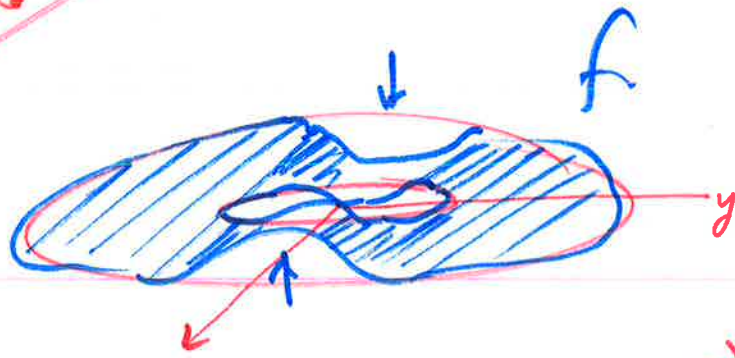
open
 $M \subset \mathbb{R}^n$, $f: M \rightarrow \mathbb{R}$

be such that
 $\text{graph}(f)$ looks like.



$$f(x, y) = \begin{cases} +\text{Bump}(y), & x > 0 \\ -\text{Bump}(y), & x \leq 0 \end{cases}$$

$$\text{Bump}(y) = e^{1/(y^2-1)} \in C^\infty$$

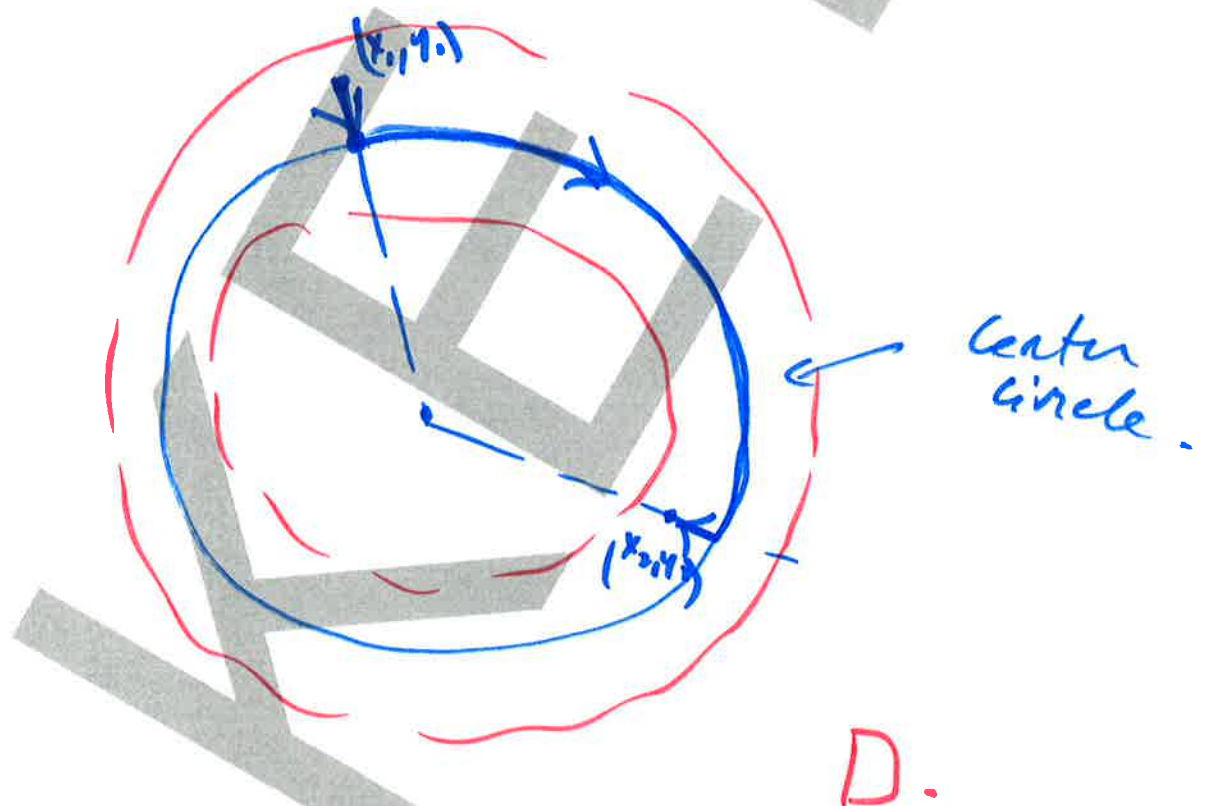


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Path connectedness of M :

$(x_1, y_1), (x_2, y_2) \in M$.



3. [3 points] Let E and F be normed spaces and $f : E \rightarrow F$ be a function. For $x \in E$ and $h \in E$, define

$$\Delta_h f(x) = f(x+h) - f(x).$$

- (a) [2 points] **Prove** or disprove the following statement:

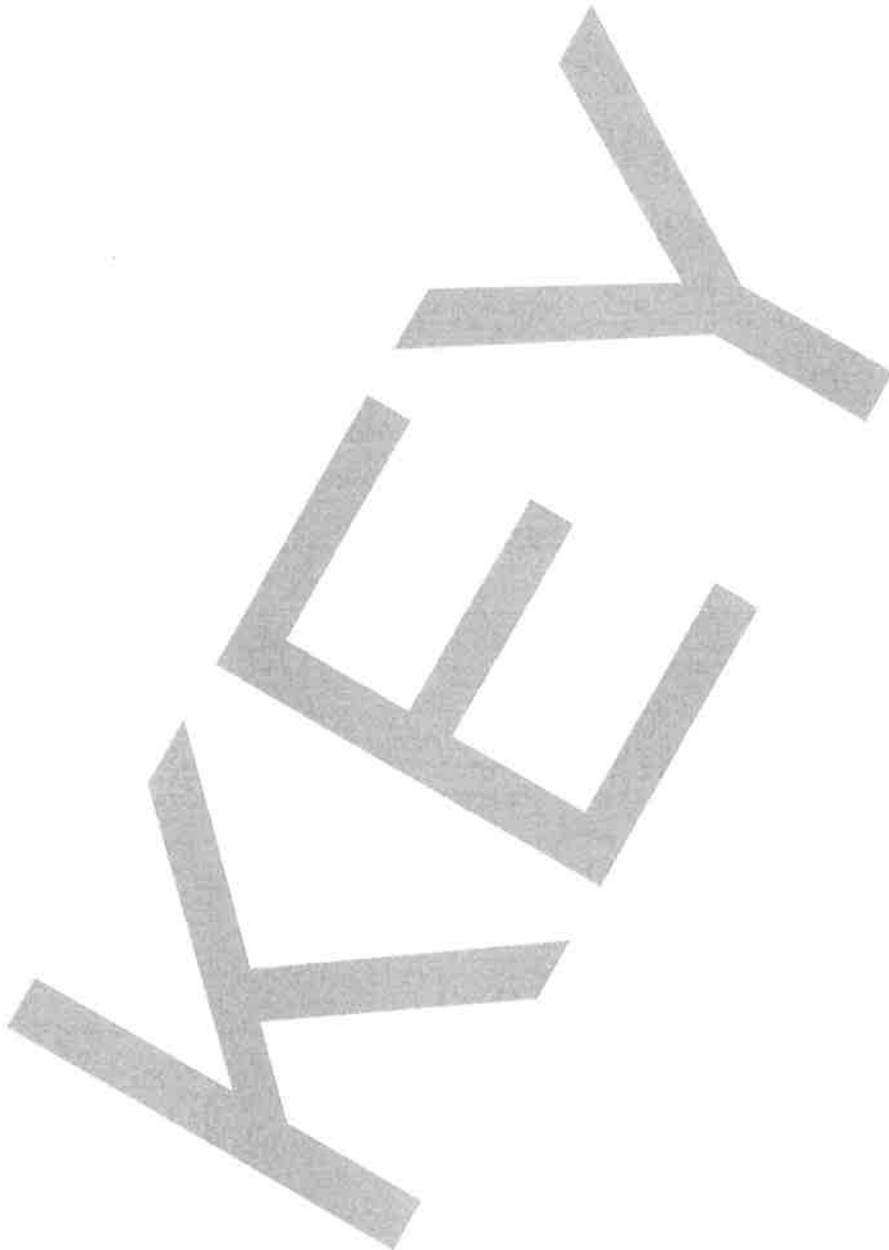
For any $x \in E$ and any $h, k \in E$,

$$\Delta_h \Delta_k f(x) = \Delta_k \Delta_h f(x).$$

$$\begin{aligned} \Delta_h \Delta_k f(x) &= \Delta_k f(x+h) - \Delta_k f(x) \\ &= f(x+k+h) - f(x+h) - f(x+k) + f(x) \leftarrow \\ \Delta_k \Delta_h f(x) &= \Delta_h f(x+k) - \Delta_h f(x) \\ &= f(x+h+k) - f(x+k) - f(x+h) + f(x) \leftarrow \end{aligned} =$$

□.

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(b) [1 point] **Prove or disprove the following statement:**

If f is twice continuously differentiable, then for any $x \in E$, there is a $\delta \in \mathbb{R}_{>0}$ such that for any $h, k \in E$, if $|h| + |k| \leq \delta$ then

$$|h| + |k| \leq \delta$$

$$|\Delta_h \Delta_k f(x)| \leq \left(\sup_{|\bar{x}-x| \leq \delta} |f''(\bar{x})|_{\text{op}} \right) |h||k|.$$

Define $\varphi: \mathbb{R}^2 \rightarrow F$

$$(s, t) \mapsto \Delta_{sh} \Delta_{tk} f(x).$$

$$\text{Then } \varphi(0, t) = \varphi(s, 0) = 0, \quad \varphi(1, 1) = \Delta_h \Delta_k f(x)$$

$$\frac{\partial \varphi}{\partial s}(s, t) = \frac{\partial}{\partial s} [f(x+sh+tk) - f(x+sh) - f(x+tk) + f(x)]$$

$$= f'(x+sh+tk)h - f'(x+sh)h$$

$$= [f'(x+sh+tk) - f'(x+sh)]h; \quad \frac{\partial \varphi}{\partial s}(s, 0) = 0.$$

$$\frac{\partial \varphi}{\partial t \partial s}(s, t) = \frac{\partial}{\partial t} ([f'(x+sh+tk) - f'(x+sh)]h)$$

$$= f''(x+sh+tk)(h, k)$$

$$\Rightarrow |\Delta_h \Delta_k f(x)|_F = |\varphi(1, 1)|_F = |\varphi(1, 1) - \varphi(0, 1)|_F$$

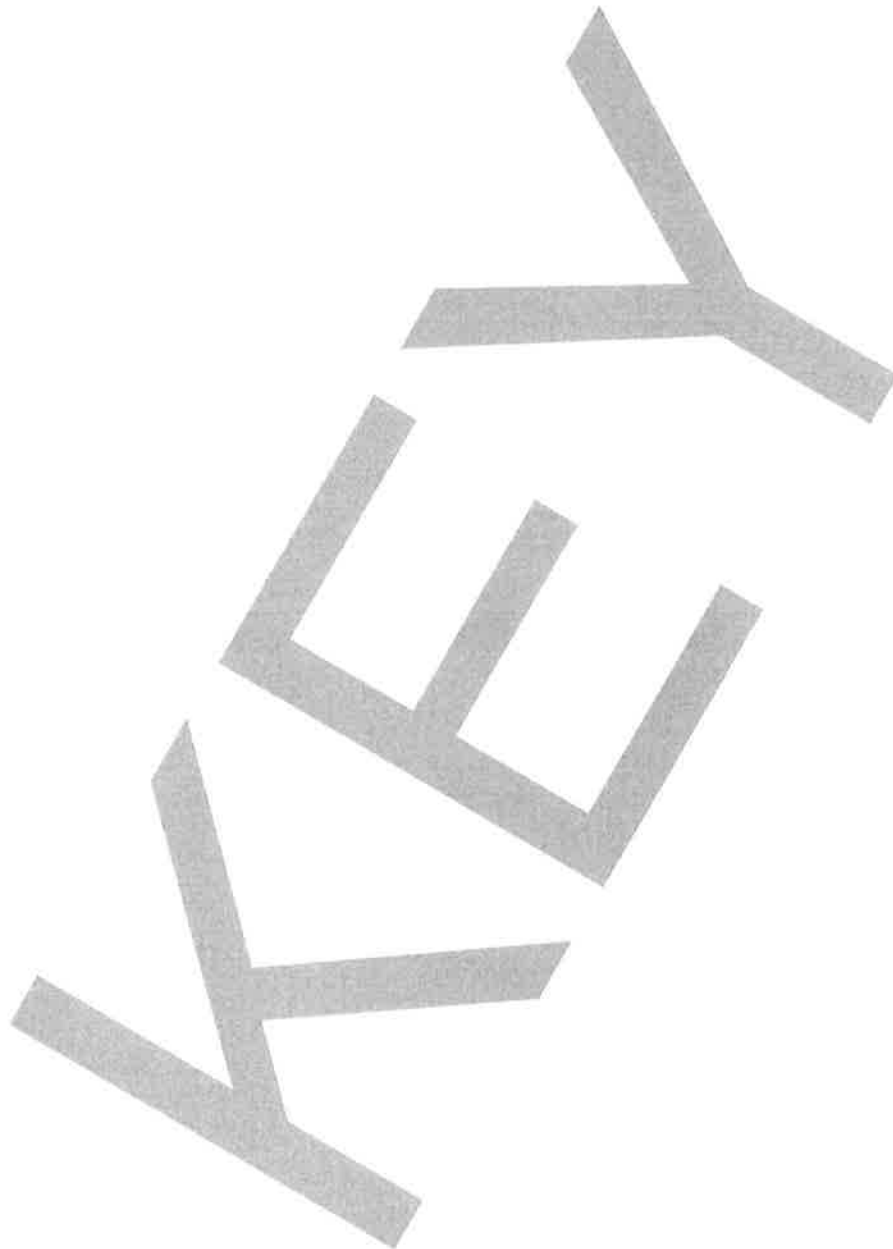
$$\leq \sup_{0 \leq s \leq 1} \left| \frac{\partial \varphi}{\partial s}(s, 1) \right|_F = \sup_{0 \leq s \leq 1} \left| \frac{\partial \varphi}{\partial s}(s, 1) - \frac{\partial \varphi}{\partial s}(s, 0) \right|_F$$

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$$\leq \sup_{0 \leq s \leq 1} \sup_{0 \leq t \leq 1} \left| \frac{\partial^2 \varphi}{\partial t \partial s}(s, t) \right|_F \leq \sup_{0 \leq s \leq 1} \sup_{0 \leq t \leq 1} |f''(x+sh+tk)| |h| |k|_E$$

□.

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